JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)
WEEK 1:
REVISION EXERCISES

Exercise Set 1: Angles and Polygons
1. Find the sum of interior angles of:
a) Pentagon
b) Octagon
c) Decagon
2. Each interior angle of a regular polygon is 140°. Find the number of sides.
3. A hexagon has angles: 110°, 120°, 125°, 115°, 130°. Find the sixth angle.
4. Each exterior angle of a regular polygon is 36°. Find:
a) Number of sides
b) Each interior angle
5. Construct:
a) An angle of 75°
b) A triangle with sides 7 cm, 6 cm, and 5 cm
c) A triangle where AB = 8 cm, angle A = 60°, and AC = 6 cm

Exercise Set 2: Angles of Elevation and Depression
1. Standing 40 m from a building, the angle of elevation to the top is 35°. Using a scale of 1 cm = 10 m, find the height of the building.
2. From a 60 m high tower, a car is seen at an angle of depression of 28°. Find the distance of the car from the tower base.
3. A ladder 12 m long leans against a wall at an angle of 65° with the ground. Find the height reached on the wall.
4. From a lighthouse 80 m high, a ship is spotted at an angle of depression of 18°. Find the distance of the ship from the lighthouse base.

Exercise Set 3: Bearings
1. Convert the following to three-figure bearings:
a) N40°E
b) S35°W
c) Northwest

2. If the bearing from A to B is 075°, find the bearing from B to A.
3. Town P is 50 km North of Q. Town R is 40 km East of Q. Using a scale of 1 cm = 10 km:
a) Find the distance PR
b) Find the bearing of R from P
4. A ship sails 60 km on a bearing of 040° from port P to point A, then 80 km on a bearing of 130° to port Q. Find:
a) The distance PQ
b) The bearing of Q from P

Exercise Set 4: Statistics – Frequency Tables
1. Create a frequency table for the following data:
Red, Blue, Red, Green, Red, Blue, Red, Green, Blue, Red, Green, Blue, Red, Red, Blue
2. The following are test scores:
56, 78, 45, 67, 56, 89, 45, 78, 56, 67, 72, 56, 89, 78, 67, 56, 45, 78, 67, 56
Create a grouped frequency table using the intervals:
40–49
50–59
60–69
70–79
80–89

Exercise Set 5: Pie Charts
1. Draw a pie chart for favorite fruits among 40 students:

Fruit	Number of Students
Apple	15
Orange	12
Banana	8
Mango	5

2. A pie chart shows the distribution of students in four clubs:

Club	Angle (degrees)
Sports	120°

Music 90°
Drama80°
Art 70°
If the total number of students is 180, find:
a) Number of students in each club
b) Which club has the most members?
c) Percentage of students in the Sports club

Exercise Set 6: Probability
1. A fair die is rolled. Find:
a) P(rolling a 4)
b) P(an even number)
c) P(a number greater than 4)
2. Two fair coins are tossed. Find:
a) P(2 Heads)
b) P(at least 1 Tail)
3. A bag contains 12 balls: 5 red, 4 blue, and 3 green. Find:
a) P(red)
b) P(not green)
c) P(blue or green)
4. If P(sunny) = 0.65, find P(not sunny).
5. A card is drawn from a standard 52-card deck. Find:
a) P(Heart)
b) P(red card)
c) P(King)

MIXED PRACTICE PROBLEMS

Problem 1
A regular polygon has each interior angle equal to 156°.

Find:
a) Number of sides
b) Each exterior angle
c) Sum of interior angles

Problem 2
From point A:
Point B is 45 km on a bearing of 125°
Point C is 60 km on a bearing of 215°
Using a scale drawing:
a) Locate points B and C
b) Find the distance BC
c) Find the bearing of C from B

Problem 3
A class of 30 students obtained the following scores:
45, 67, 52, 78, 56, 67, 89, 45, 72, 56, 61, 78, 67, 53, 89, 72, 56, 45, 67, 78, 55, 82, 48, 63, 71, 85, 59, 74, 68, 51
a) Create a grouped frequency table (use intervals of your choice)
b) Draw a pie chart
c) What percentage of students scored between 70 and 79?

Problem 4
Two fair dice are rolled. Find:
a) P(sum = 8)
b) P(doubles)
c) P(sum greater than 9)

Problem 5 – Triangle Construction
Given: AB = 9 cm, angle A = 50°, angle B = 60°
a) Construct triangle ABC

b) Measure sides AC and BC
c) Bisect angle C
d) What is the measure of angle C?